

Graph Theory & N/w Egⁿ

Unit - 1

1. Explain the following with respect to n/w topology

- i) Planar and Non-Planar graph
- ii) Incidence matrix
- iii) Number of possible trees

Solution:-

i) Planar graph:- It's a graph drawn on a plane surface such that no branch passes over or under any other branch.

Non-planar graph:- It is a graph that can't be drawn on a plane without any branches crossing each other.

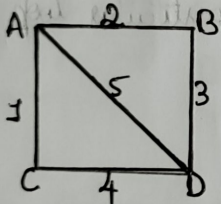


Fig: Planar graph

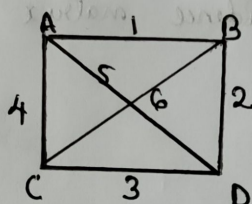


Fig:- Non-Planar graph

ii) Incidence Matrix

The Incidence matrix of a particular graph has 2 types reduced and complete. The Reduced incidence matrix A is obtained from a complete incidence matrix A_{ij} by eliminating one of the rows it is of the order $(n-1) \times j$.

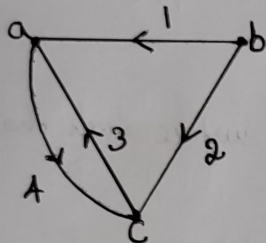
The Complete incidence matrix / all incidence matrix

It is a matrix which gives complete information regarding the connections of various branches to the nodes and the orientation of these branches towards the nodes.

Procedure to write incidence matrix is as follows

- The number of branches are indicated in a row
- The nodes are indicated as a column
- If a particular branch is not connected to a particular node, the corresponding element in the matrix table is filled zero.
- If a particular branch is connected to a particular node and its orientation is away from the node, the corresponding element is filled as +1 and the orientation is towards the node then the matrix table is filled with -1.

Example:-



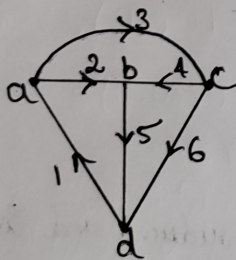
nodes	Branches			
	1	2	3	4
a	-1	0	-1	1
b	1	1	0	0
c	0	-1	1	-1

$$A_{ij} = \begin{bmatrix} -1 & 0 & -1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

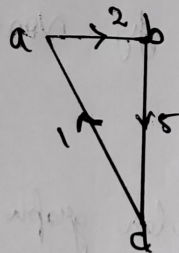
→ It is seen from the complete incidence matrix A_{ij} that sum of all the elements in any column is zero.

→ The determinants of the incidence matrix for a particular loop called as loop matrix is always zero.

Example:-



⇒

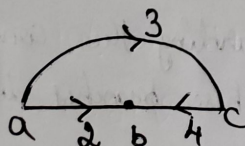


⇒

$$A_{ij} = \begin{bmatrix} 1 & 2 & 5 \\ a & -1 & 1 & 0 \\ b & 0 & -1 & 1 \\ d & 1 & 0 & -1 \end{bmatrix}$$

$$\text{Determinants} \Rightarrow -1(1-0) - 1(0-1) + 0$$

$$\begin{aligned} &= -1 + 1 \\ &= 0 \end{aligned}$$



$$\Rightarrow A_{ij} = \begin{bmatrix} 2 & 3 & 4 \\ a & 1 & 1 & 0 \\ b & -1 & 0 & -1 \\ c & 0 & -1 & 1 \end{bmatrix}$$

$$\text{Determinants} \Rightarrow (1(0) - 1(-1 \times 1)) - 1(-1 \times 1) + 0$$

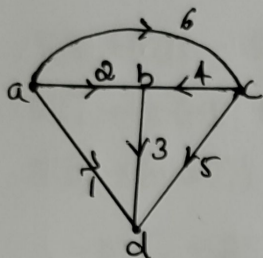
$$\begin{aligned} &= -1 + 1 \\ &= 0 \end{aligned}$$

Number of possible trees

(3)

A tree is a connected subgraph of a n/w that contains all the nodes but does not contain any closed path or loop. The number of different spanning trees that can be formed from a network graph.

Example:-



Nodes	Branches					
	1	2	3	4	5	6
a	1	1	0	0	0	1
b	0	-1	1	-1	0	0
c	0	0	0	1	1	-1
d	-1	0	-1	0	-1	0

$$A_{ij} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \\ -1 & 0 & -1 & 0 & -1 & 0 \end{bmatrix}$$

Reduced Incidence Matrix

$$A_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 & -1 & 0 \end{bmatrix} \quad \text{Remove row 'c'}$$

3x6

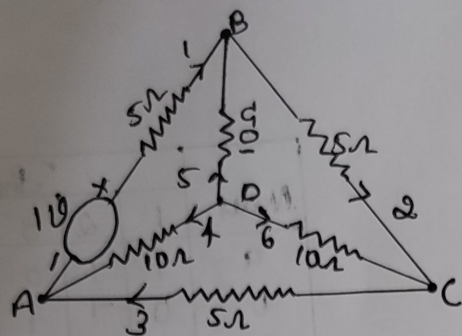
$$\text{NO of possible trees} = |AA^T|$$

$$A^T = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \quad 6 \times 3$$

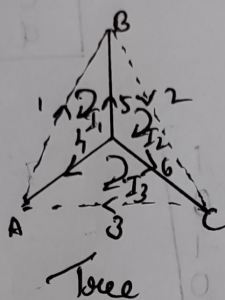
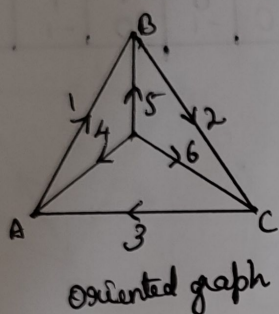
$$AA^T = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

$$[AA^T] = 3(9-1) + 1(-3-1) - 1(1+3) = 16 //$$

2. For the n/w shown in fig, write the tie set schedule, and obtain equation using KVL. Calculate loop currents. Follow the same orientation and numbering as shown in fig. Use branch 4, 5, 6 as tree branches



Solution:-
⇒



Tie set 1: {1, 4, 5}
Tie set 2: {2, 5, 6}
Tie set 3: {3, 4, 6}

Tie set schedule

	1	2	3	4	5	6
$I_1(1, 4, 5)$	1	0	0	1	-1	0
$I_2(2, 5, 6)$	0	1	0	0	1	-1
$I_3(3, 4, 6)$	0	0	1	-1	0	1

Row wise

$$V_1 + V_4 - V_5 = 0 \rightarrow (A)$$

$$V_2 + V_5 - V_6 = 0 \rightarrow (B)$$

$$V_3 - V_4 + V_6 = 0 \rightarrow (C)$$

Column wise

$$I_1 = I_1$$

$$I_2 = I_2$$

$$I_3 = I_3$$

$$I_4 = I_1 - I_3$$

$$I_5 = -I_1 + I_2$$

$$I_6 = -I_2 + I_3$$

Resistor

$$R_1 = R_2 = R_3 = 5\Omega$$

$$R_4 = R_5 = R_6 = 10\Omega$$

Using Ohm's law

$$V_1 = 5I_1$$

$$V_5 = 10I_5$$

$$V_2 = 5I_2$$

$$V_6 = 10I_6$$

$$V_3 = 5I_3$$

$$V_4 = 10I_4$$

$$V_1 = 5(I_1) - 1$$

$$V_6 = 10(-I_2 + I_3)$$

$$V_2 = 5I_2$$

$$V_3 = 5I_3$$

$$V_4 = 10(I_1 - I_3)$$

$$V_5 = 10(-I_1 + I_2)$$

⑤

① $\Rightarrow$

$$5j_1 - 1 + 10j_4 - 10j_5 = 0$$

$$5I_1 - 1 + 10(I_1 - I_3) - 10(-I_1 + I_2) = 0$$

$$5I_1 - 1 + 10I_1 - 10I_3 + 10I_1 - 10I_2 = 0$$

$$25I_1 - 10I_2 - 10I_3 = 1 \rightarrow \textcircled{1}$$

② $\Rightarrow$

$$5j_2 + 10j_5 - 10j_6 = 0$$

$$5I_2 + 10(-I_1 + I_2) - 10(-I_2 + I_3) = 0$$

$$5I_2 + (-10I_1) + 10I_2 + 10I_2 - 10I_3 = 0$$

$$-10I_1 + 25I_2 - 10I_3 = 0 \rightarrow \textcircled{2}$$

③ $\Rightarrow$

$$5j_3 - 10j_4 + 10j_6 = 0$$

$$5I_3 - 10(I_1 - I_3) + 10(-I_2 + I_3) = 0$$

$$5I_3 - 10I_1 + 10I_3 + (-10I_2) + 10I_3 = 0$$

$$-10I_1 - 10I_2 + 25I_3 = 0 \rightarrow \textcircled{3}$$

$$I_1 = 0.085A$$

$$I_2 = 0.057A$$

$$I_3 = 0.057A$$

$\Rightarrow V_1 = 5(I_1) - 1$

$$= 25(0.085) - 1$$

$$= -0.575V$$

$V_2 = 5(0.057)$

$$= 0.285V$$

$V_3 = 5(0.057)$

$$= 0.285V$$

$V_4 = 10(0.085 - 0.057)$

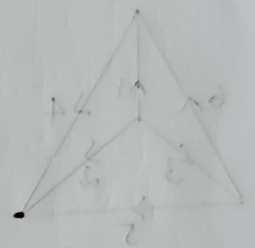
$$= 0.28V$$

$V_5 = 10(-0.085 + 0.057)$

$$= -0.28V$$

$V_6 = 10(-0.057 + 0.057)$

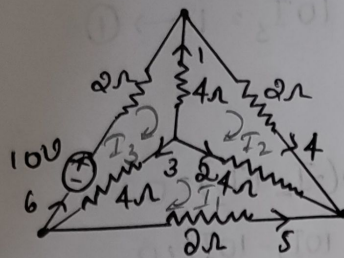
$$= 0V$$



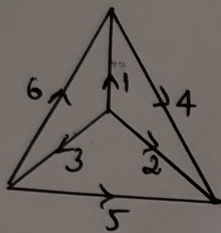
Node	1	2	3	4	5	6
1	0	1	0	1	1	0
2	0	0	1	0	1	1
3	1	0	0	1	0	1

① $\left\{ \begin{array}{l} 0 = 2V - V - 5V \\ 0 = 1V + 2V - V \\ 0 = 2V + V + 1V \end{array} \right.$

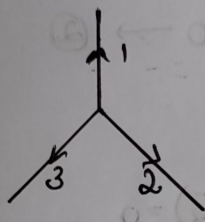
3. Draw the oriented graph of a network shown in figure. Select a tree write the tie set schedule and obtain equilibrium equation



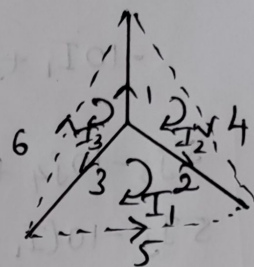
Solution:-



oriented graph



Tree



Loop currents	Branches					
	1	2	3	4	5	6
$I_1(2,3,5)$	0	1	-1	0	-1	0
$I_2(1,2,4)$	1	-1	0	1	0	0
$I_3(1,3,6)$	-1	0	1	0	0	1

Row wise:-

$$\left. \begin{aligned} V_2 - V_3 - V_5 &= 0 \\ V_1 - V_2 + V_4 &= 0 \\ -V_1 + V_3 + V_6 &= 0 \end{aligned} \right\} \rightarrow \text{①}$$

Column wise:-

$$\begin{aligned} I_1 &= I_2 - I_3 & I_3 &= -I_1 + I_3 & I_5 &= -I_1 \\ I_2 &= I_1 - I_2 & I_4 &= I_2 & I_6 &= I_3 \end{aligned}$$

Given :- $R_1 = 4\Omega, R_2 = 4\Omega, R_3 = 4\Omega, R_4 = 2\Omega, R_5 = 2\Omega, R_6 = 2\Omega$

(7)

By ohm's law

$$\begin{aligned}
 V_1 &= I_1 R_1 & V_3 &= I_3 R_3 & V_5 &= I_5 R_5 \\
 &= 4(I_2 - I_3) & &= 4(-I_1 + I_3) & &= -2I_1 \\
 V_2 &= I_2 R_2 & V_4 &= I_4 R_4 & V_6 &= I_6 R_6 \\
 &= 4(I_1 - I_2) & &= 2I_2 & &= 2I_3 - 10
 \end{aligned}
 \left. \vphantom{\begin{aligned} V_1 \\ V_2 \end{aligned}} \right\} \rightarrow (2)$$

Substituting (2) in (1)

$$V_2 - V_3 - V_5 = 0$$

$$4(I_1 - I_2) - (-I_1 + I_3) - (-2I_1) = 0$$

$$4I_1 - 4I_2 + I_1 - I_3 + 2I_1 = 0$$

$$10I_1 - 4I_2 - I_3 = 0 \rightarrow (3)$$

$$V_1 - V_2 + V_4 = 0$$

$$4(I_2 - I_3) - 4(I_1 - I_2) + 2I_2 = 0$$

$$4I_2 - 4I_3 - 4I_1 + 4I_2 + 2I_2 = 0$$

$$-4I_1 + 10I_2 - 4I_3 = 0 \rightarrow (4)$$

$$-V_1 + V_3 + V_6 = 0$$

$$-4(I_2 - I_3) + 4(-I_1 + I_3) + 2I_3 - 10 = 0 \Rightarrow I_1 = 1.42A$$

$$-4I_2 + 4I_3 + (-4I_1) + 4I_3 + 2I_3 = 10 \Rightarrow I_2 = 1.42A$$

$$-4I_1 - 4I_2 + 10I_3 = 10 \rightarrow (5) \Rightarrow I_3 = 2.14A$$

Branch current:-

$$I_1 = I_2 - I_3$$

$$= 1.42 - 2.14 = -0.72A$$

$$I_2 = I_1 - I_2 = 0A$$

$$I_3 = -I_1 + I_3 = 0.72A$$

$$I_4 = I_2 = 1.42A$$

$$I_5 = -I_1 = -1.42A$$

$$I_6 = I_3 = 2.14A$$

Branch voltage:-

$$V_1 = 4(I_2 - I_3) \Rightarrow 4 \times 0.72 = -2.88V$$

$$V_2 = 4(I_1 - I_2) \Rightarrow 4 \times 0 = 0V$$

$$V_3 = 4(-I_1 + I_3) \Rightarrow 4 \times 0.72 = 2.88V$$

$$V_4 = 2I_2 \Rightarrow 2 \times 1.42 = 2.84V$$

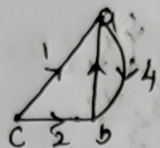
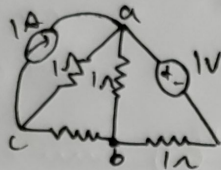
$$V_5 = -2I_1 \Rightarrow -2 \times 1.42 = -2.84V$$

$$V_6 = 2I_3 - 10 \Rightarrow 2(2.14) - 10 = -5.72V$$

4)

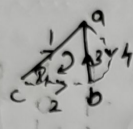
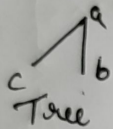


1Ω resistor in parallel with 1V source redundant



1V is in series with 1Ω

oriented graph



Tree-Set Schedule

Tree-Set	1	2	3	4
$T_1(1,2,3)$	1	-1	1	0
$T_2(3,4)$	0	0	1	1

voltage Equations:- $V_1 - V_2 - V_3 = 0$
 $V_3 + V_4 = 0 \rightarrow \textcircled{a}$

branch current Eq:- $i_1 = I_1, i_2 = -I_1, i_3 = -I_1 + I_2, i_4 = I_2$

Resistances:- $R_1 = R_2 = R_3 = R_4 = 1\Omega$

Branch voltage:- $V_1 = i_1 R_1 = 1, V_2 = i_2 R_2, V_3 = i_3 R_3, V_4 = i_4 R_4 + 1$
 $V_1 = I_1 - 1, V_2 = -I_1, V_3 = (-I_1 + I_2), V_4 = I_2 + 1$

Sub in $\textcircled{a}$

$$\begin{aligned} V_1 - V_2 - V_3 &= 0 \\ I_1 - 1 + I_1 + I_1 - I_2 &= 0 \\ 3I_1 - I_2 &= 1 \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} V_3 + V_4 &= 0 \\ -I_1 + I_2 + I_2 + 1 &= 0 \\ 2I_2 - I_1 + 1 &= 0 \\ -I_1 + 2I_2 &= -1 \rightarrow \textcircled{2} \end{aligned}$$

$$I_1 = 0.2A$$

$$I_2 = -0.4A$$

Branch currents

$$i_1 = 0.2A$$

$$i_2 = -0.2A$$

$$i_3 = -0.6A$$

$$i_4 = -0.4A$$

Branch voltages

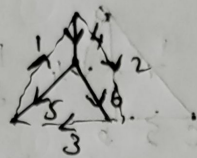
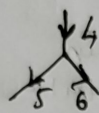
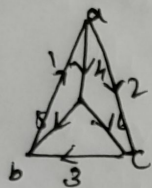
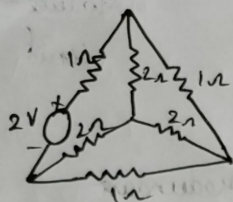
$$V_1 = -0.8V$$

$$V_2 = -0.2V$$

$$V_3 = -0.6V$$

$$V_4 = 0.6V$$

5)



	1	2	3	4	5	6
$I_1(1,4,5)$	1	0	0	1	1	0
$I_2(2,4,6)$	0	1	0	-1	0	-1
$I_3(3,5,6)$	0	0	1	0	-1	1

$$B_2 = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

Column will

$$I_1 = I_1$$

$$I_2 = I_2$$

$$I_3 = I_3$$

$$I_4 = I_1 - I_2$$

$$I_5 = I_1 - I_3$$

$$I_6 = -I_2 + I_3$$

Row will

$$V_1 + V_4 + V_5 = 0 \rightarrow (A)$$

$$V_2 - V_4 - V_6 = 0 \rightarrow (B)$$

$$V_3 - V_5 + V_6 = 0 \rightarrow (C)$$

using ohm's law

$$V_1 = I_1 - 2$$

$$V_2 = I_2$$

$$V_3 = I_3$$

$$V_4 = 2(I_1 - I_2)$$

$$V_5 = 2(I_1 - I_3)$$

$$V_6 = 2(-I_2 + I_3)$$

$$(A) \Rightarrow I_1 - 2 + 2I_1 - 2I_2 + 2I_1 - 2I_3 = 0$$

$$5I_1 - 2I_2 - 2I_3 = 2 \rightarrow (1)$$

$$(B) \Rightarrow I_2 - 2I_1 - 2I_2 + 2I_2 - 2I_3 = 0$$

$$-2I_1 + 5I_2 - 2I_3 = 0 \rightarrow (2)$$

$$(C) \Rightarrow I_3 - 2I_1 - 2I_3 + 2I_2 - 2I_3 = 0$$

$$-2I_1 - 2I_2 + 5I_3 = 0 \rightarrow (3)$$

$$I_1 = 0.85A$$

$$I_2 = 0.57A$$

$$I_3 = 0.57A$$

$$I_1 = 0.85A$$

$$I_2 = 0.57A$$

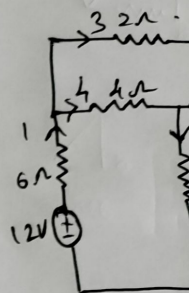
$$I_3 = 0.57A$$

$$I_4 = 0.28A$$

$$I_5 = 0.28A$$

$$I_6 = 0A$$

6)



Loop will

$$I_1(3,4,5)$$

$$I_2(1,4,6)$$

$$I_3(2,5,6)$$

Row will

$$V_3 - V_4$$

$$V_1 + V_2$$

$$V_2 +$$

overdetermined

R_1

By ohm's law

$$V_1 =$$

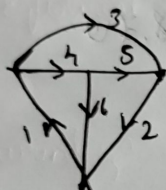
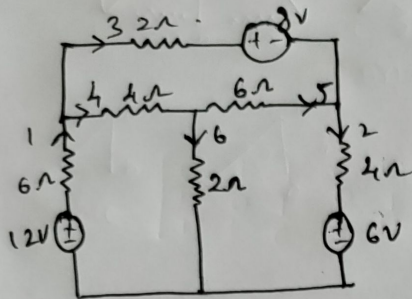
$$V_2 =$$

$$V_3 =$$

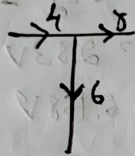
$$\begin{aligned} I_1 &= 0.85 \text{ A} \\ I_2 &= 0.57 \text{ A} \\ I_3 &= 0.57 \text{ A} \\ I_4 &= 0.28 \text{ A} \\ I_5 &= 0.28 \text{ A} \\ I_6 &= 0 \text{ A} \end{aligned}$$

$$\begin{aligned} V_1 &= -1.15 \text{ V} \\ V_2 &= 0.57 \text{ V} \\ V_3 &= 0.57 \text{ V} \\ V_4 &= 0.57 \text{ V} \\ V_5 &= 0.57 \text{ V} \\ V_6 &= 0 \text{ V} \end{aligned}$$

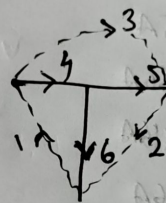
6)



oriented graph



Tree



loop current	1	2	3	4	5	6
$I_1(3, 4, 5)$	0	0	1	-1	-1	0
$I_2(1, 4, 6)$	1	0	0	1	0	1
$I_3(2, 5, 6)$	0	1	0	0	1	-1

Row will

$$\left. \begin{aligned} V_3 - V_4 - V_5 &= 0 \\ V_1 + V_4 + V_6 &= 0 \\ V_2 + V_5 - V_6 &= 0 \end{aligned} \right\} \rightarrow \text{a)}$$

column will

$$\begin{aligned} I_1 &= I_2 \\ I_2 &= I_3 \\ I_3 &= I_1 \\ I_4 &= -I_1 + I_2 \\ I_5 &= -I_1 + I_3 \\ I_6 &= I_2 - I_3 \end{aligned}$$

consider :

$$R_1 = 6\Omega, R_2 = 4\Omega, R_3 = 2\Omega, R_4 = 4\Omega, R_5 = 6\Omega, R_6 = 2\Omega$$

By ohm's law

$$\begin{aligned} V_1 &= 6I_2 - 12 \\ V_2 &= 4I_3 + 6 \\ V_3 &= 2I_1 + 8 \end{aligned}$$

$$\left. \begin{aligned} V_4 &= 4(-I_1 + I_2) \\ V_5 &= 6(-I_1 + I_3) \\ V_6 &= 2(I_2 - I_3) \end{aligned} \right\} \rightarrow \text{b)}$$

Sub ② in ①

$$V_3 - V_4 - V_6 = 0$$

$$12I_1 - 4I_2 - 6I_3 = -8 \rightarrow ①$$

$$V_1 + V_4 + V_6 = 0$$

$$-4I_1 + 12I_2 - 2I_3 = 12 \rightarrow ②$$

$$V_2 + V_5 - V_6 = 0$$

$$-6I_1 - 2I_2 + 12I_3 = -6 \rightarrow ③$$

$$I_1 = -0.916A$$

$$I_2 = 0.55A$$

$$I_3 = -0.866A$$

Branch currents

$$I_1 = 0.55A$$

$$I_2 = -0.866A$$

$$I_3 = -0.916A$$

$$I_4 = 1.466A$$

$$I_5 = 0.05A$$

$$I_6 = 1.416A$$

Branch voltage

$$V_1 = 6I_2 - I_2 = -8.7V$$

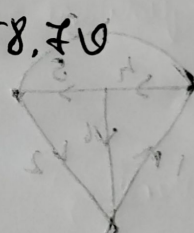
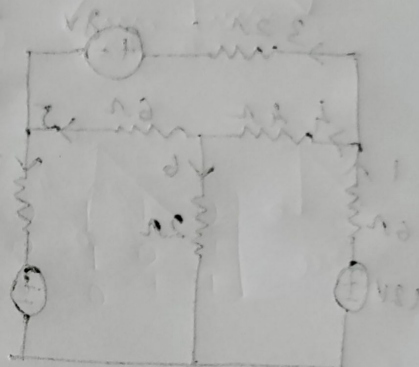
$$V_2 = 2.53V$$

$$V_3 = 6.168V$$

$$V_4 = 5.864V$$

$$V_5 = 0.3V$$

$$V_6 = 2.83V$$



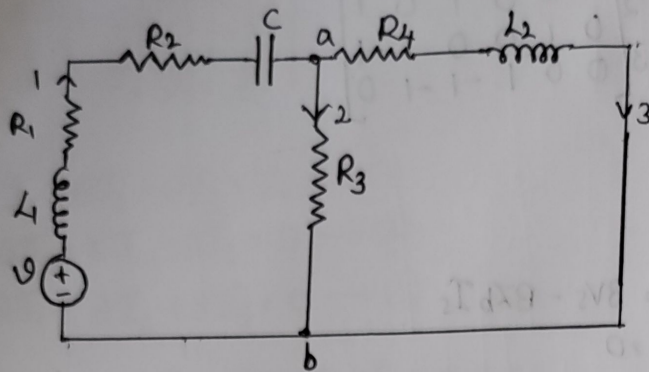
	1	2	3	4	5	6	known
1	0	1	1	0	0	0	(3, 4, 5), 1
2	1	0	1	0	0	1	(2, 4, 1), 2
3	1	1	0	0	1	0	(2, 3, 6), 3

the results

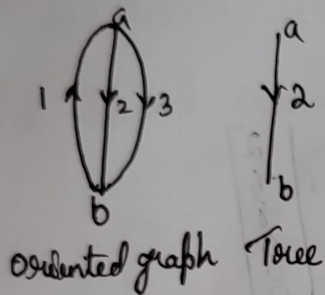
the results

$$\begin{aligned} 0 &= 2V - 4V - 6V \\ 0 &= 4V + 4V + 4V \\ 0 &= 4V - 2V + 4V \end{aligned}$$

4) For the n/w shown in figure, write the incidence matrix tree



Solution:-

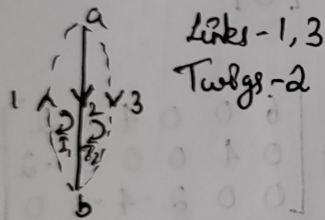


Incidence matrix

nodes	1	2	3
a	-1	1	1
b	1	-1	-1

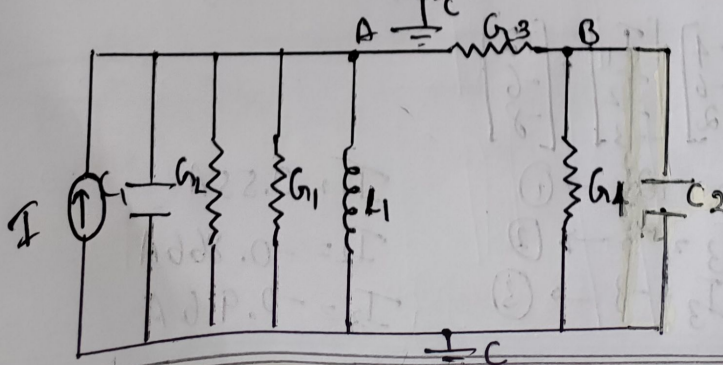
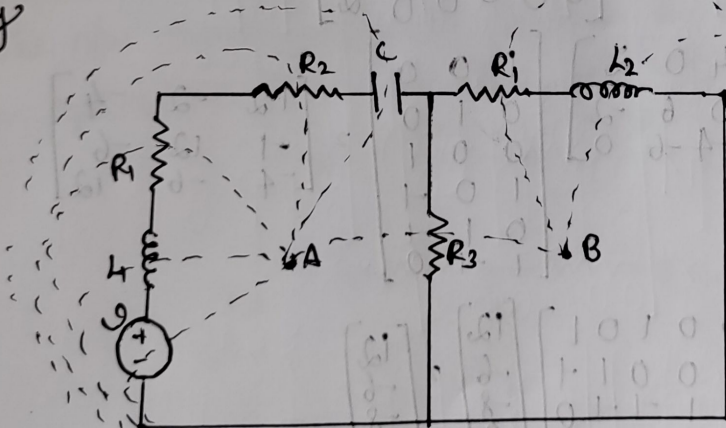
Complete incidence matrix

$$A_{ij} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \end{bmatrix}$$

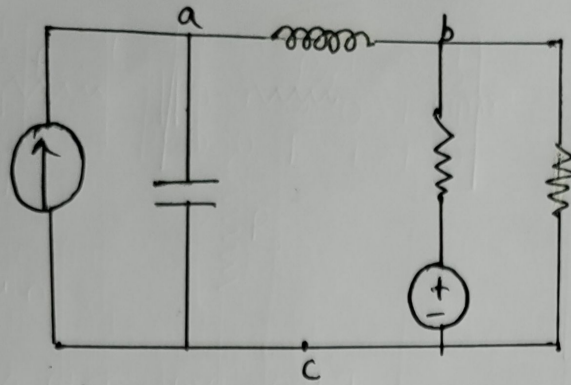


$$B_2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

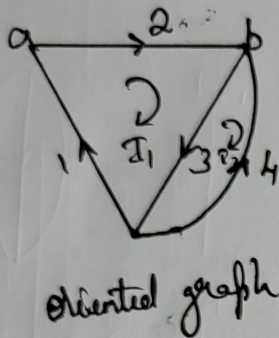
Duality



Find the maximum possible number of trees for the net shown in fig (15)



Solution:-
⇒



nodes	1	2	3	4
a	-1	1	0	0
b	0	-1	1	-1
c	1	0	-1	1

$$A_1 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix}$$

Complete incidence matrix

$$A_2 = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

Reduced incidence matrix

Maximum possible number of trees = $[AA^T]$

$$AA^T = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

$$|AA^T| = 6 - 1$$

$$= 5$$